

Table of Contents

1. Topic introduction	3
1.1 Source	3
1.2 Background	3
1.3 Significancy	5
2. Related Work.....	7
Section 1: Machine Learning and reinforcement learning Algorithm based Drone system	8
Section 2: A-Star Algorithm for drone system	13
Section 3: Search and Detection Operation in drone system	15
3. Research Program.....	17
3.1 Research Objective and content.....	18
3.2 Key Research Problems.....	19
3.3 Methods	19
3.4 Novelty	23
4. Expected Objectives and Achievements	23
1. Expected research objectives:	23
2. Expected Achievements:	27
5. Research plan.....	28
6. References.....	29

1. Topic introduction

1.1 Source

This research implements a comprehensive approach to drone path planning and optimization within a simulated environment, leveraging a combination of the A* algorithm and reinforcement learning (RL). The primary objective is to develop a robust framework allowing the drone to autonomously navigate complex terrains with static and dynamic obstacles to reach its target efficiently in urban search and rescue missions.

The environment is simulated as a 2D grid using Python, where obstacles, targets, and environmental dynamics are defined. The A* algorithm provides an initial optimal path for the drone based on static obstacles, ensuring an efficient and collision-free trajectory. To enhance adaptability in dynamic environments, reinforcement-learning is integrated into the system. A deep Q-Network (DQN) is utilized to train the drone to adjust its path in real time, optimizing its movements based on a reward system that encourages target proximity and penalizes collisions or inefficient routes.

The simulation is visualized using Python libraries, including Pygame for environment representation and Matplotlib for monitoring performance metrics such as path efficiency, collision rates, and learning progression. The results demonstrate that integrating reinforcement learning techniques with classical path planning algorithms significantly improves navigation performance, especially in dynamic and unpredictable scenarios. This research lays a foundation for developing intelligent autonomous systems for search and rescue, delivery services, and other real-world applications.

1.2 Background

Drones have become transformative tools in various civilian and emergency response applications. Their capabilities enormously enhance the efficiency of operations, demonstrating their potential to contribute positively to society. These autonomous or remotely piloted aircraft systems offer mobility, flexibility, and aerial perspective capabilities, making them particularly valuable in time-critical scenarios such as search and rescue (SAR) operations. The integration of drones in SAR operations has shown remarkable potential to enhance the efficiency and

effectiveness of emergency response efforts, especially in scenarios where traditional ground-based search methods face significant limitations or risks that the drones cannot be affected by. Search and rescue missions are challenging and time-sensitive missions aimed at finding and rescuing people in distress or imminent danger. The conventional approach to SAR operations, which primarily relies on ground teams and human-crewed aircraft, often faces challenges such as limited visibility, difficult terrain accessibility, and significant resource requirements regarding personnel and equipment.

Through their unique characteristics, drones present a compelling solution to many challenges. Their ability to rapidly deploy, operate in challenging environments, provide real-time aerial surveillance, and operate in a timely manner makes them invaluable assets in SAR missions. Moreover, the reasonably low operational costs and reduced risk to human personnel make drones attractive for emergency response organizations. However, effectively utilizing drones in SAR operations requires educated path planning and optimization capabilities to maximize their potential impact.

Path planning, in the context of drone operations, refers to locating and determining the optimal route for a drone to navigate through from its current position to a target location while considering various constraints and objectives. These constraints typically include obstacle avoidance, battery limitations, and environmental conditions which affect the mission's success. In SAR scenarios, effective path planning becomes particularly crucial as it directly impacts the probability of successfully locating the operations targets in the shortest possible time while managing limited resources efficiently.

The complexity of drone path planning and optimization in SAR missions is heightened by several factors:

1. Real time Decision making: The need to process and respond to dynamic environmental information and changing mission parameters in real-time
2. resource optimization and balancing: managing limited battery life while maximizing search coverage
3. Environmental challenges: navigating through complex urban or natural environments with various obstacles.

4. target uncertainty: operating with incomplete or uncertain information about the target precise location.
5. time Criticality: balancing thorough search patterns with the urgency of the mission

Traditional path-planning approaches often struggle to address these challenges systematically, particularly in complex urban environments with numerous obstacles and highly constrained search space. While existing research has made significant progress in individual aspects of drone navigation, there remains a crucial need for integrated solutions that can handle real-time path planning while optimizing it.

Recent achievements in artificial intelligence and autonomous systems have opened new possibilities for enhancing drone capabilities in SAR operations. Particularly, developments in areas such as reinforcement learning and computer vision have shown promising results in improving autonomy and effectiveness of drone operations. However, the integration of these technologies and methods into a cohesive system capable of real-time operation in complex environments remains a significant challenge.

This research addresses these challenges by developing an intelligent real-time path planning and optimization system designed explicitly for drone-based SAR missions.

The proposed system aims to improve SAR operations Through:

- Real-time path planning with dynamic obstacle avoidance capabilities.
- battery-efficient routing algorithms that maximize search coverage in the targeted area.
- adaptive navigation strategies for complex urban environments.

1.3 Significancy

The deployment of drones has become increasingly essential in both military and civilian applications:

1. Military Applications:

- Risk Reduction: Drones eliminate the need to risk human personnel in dangerous reconnaissance and surveillance missions. [1],[2]
- Cost Effectiveness: Unmanned systems are significantly less expensive to deploy and maintain compared to manned aircraft. [3],[4]

- **Enhanced Capabilities:** Drones can operate in hazardous environments, maintain extended surveillance, and access areas that would be impossible or too risky for human operators. [4]
- **Tactical Advantage:** Ability to gather real-time intelligence without detection and operate in hostile territories with minimal risk [4]

2. Civilian Applications:

- **Disaster Response:** Rapid deployment for assessment and aid in natural disasters [4],[2]
- **Infrastructure Inspection:** Safe and efficient inspection of bridges, power lines, and tall structures
- **Agricultural Monitoring:** Precision agriculture and crop monitoring
- **Emergency Services:** Support for firefighting, law enforcement, and search and rescue operations
- **Environmental Protection:** Wildlife monitoring and environmental assessment [2]
- **Urban Planning:** Aerial mapping and urban development monitoring [2],[5]

The importance of drones in search and rescue (SAR) missions is demonstrated through several critical factors:

1. Enhanced Search Capabilities

- **Rapid Coverage:** Drones can quickly survey large areas that would take ground teams hours or days to cover. [1],[2]
- **Access to Remote Areas:** Ability to reach and search areas that are difficult or dangerous for ground teams to access. [4]
- **Aerial Perspective:** Provides valuable overhead views that can reveal clues or patterns not visible from ground level. [5]

2. Time-Critical Response

- **Quick Deployment:** Drones can be launched within minutes of arrival at a search site. [3],[5]
- **Real-time Information:** Immediate transmission of visual data to rescue teams. [2],[4]
- **Extended Operation:** Ability to maintain continuous search operations in suitable weather conditions. [4],[3]

3. Resource Optimization

- **Cost-Effective:** Reduced operational costs compared to manned aircraft. [3]

- **Team Efficiency:** Better allocation of ground search teams based on drone observation. [3], [4]
- **Risk Reduction:** reduce the risk of saving people in hard conditions. [2]

4. Search Effectiveness

- **Multi-sensor Capability:** Drones Can be equipped with a lot of sensors (thermal, optical, etc.) for various search conditions. [5]
- **Systematic Coverage:** Ensures methodical search patterns with maximum coverage. [3]
- **Documentation:** Automatic recording of searched areas and findings. [2]

2. Related Work

Approach	How It Works	Advantages	Disadvantages
A* Algorithm*	Uses a heuristic function to calculate the shortest path from the starting point to the goal based on the cost function.	- Efficient for static environments	- Poor performance in dynamic environments
Dijkstra's Algorithm	Explores all possible paths and selects the shortest one based on cost.	- Guarantees shortest path	- Computationally expensive
		- Suitable for simple environments	- Slow for large environments
Rapidly-exploring Random Tree (RRT)	Expands random paths towards the target and refines them as needed.	- Fast for pathfinding in complex environments	- Inefficient in narrow spaces
		- Suitable for non-uniform maps	- May not always find the shortest path
Potential Field Method	Treats the goal as an attracting force and obstacles as repelling forces, generating a potential field for movement.	- Easy to implement	- Stuck in local minima
My Approach (Reinforcement Learning)	Trains a model using trial-and-error (reward-based) learning to adapt and find the optimal path based on real-time environmental feedback.	- Adaptable to dynamic environments	- Requires extensive training data

approaches comparison

Section 1: Machine Learning and reinforcement learning Algorithm based Drone system

1.1 Real time 3D Path Planning for UAV Navigation

Tullu et al. [6] proposed a machine-learning approach to real-time 3D path planning for autonomous UAV navigation in cluttered environments. The authors approach involved integrating YOLO (You Only Look Once) object detection algorithm with a lightweight 3D path planner to enable real-time obstacle avoidance. Their approach differs from traditional path planning algorithms like A* by avoiding exhaustive node exploration and instead mapping the UAV's current location to select nodes between consecutive obstacles. The system uses a ZED mini stereo camera for object detection and localization, combined with GPS and LiDAR for UAV positioning. The path planner considers both open obstacles (requiring passage through, like windows) and closed obstacles (requiring avoidance maneuvers), applying a heuristic function for cost minimization. The effectiveness of their approach was through Software-In-The-Loop (SITL) simulations tests in Gazebo environments and preliminary real-world flight tests. Results demonstrated successful autonomous navigation through various obstacles, including pedestrians, windows, poles, tunnels, and trees. An essential contribution is decreasing the computational burden on the companion computer by integrating fast object detection with efficient path planning, making it suitable for real-time applications. However, limitations included GPS accuracy issues in open fields and the need for further real-world testing in more complex environments.

1.2: Data Analysis and Modeling for Drone Communications using ML algorithms

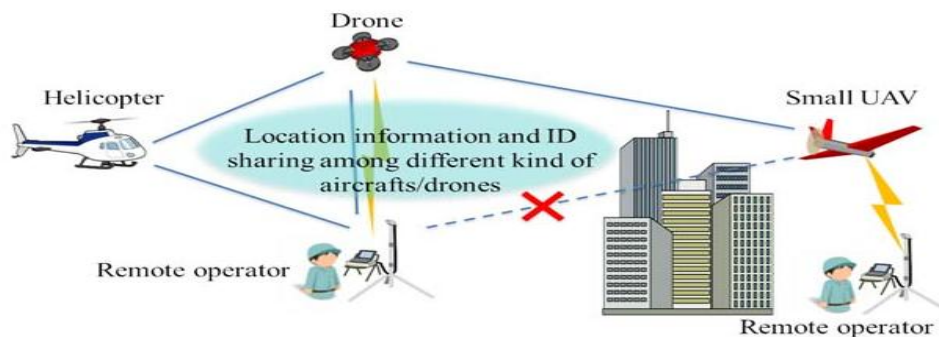


Figure 1: Concept of drone location and id Information sharing system

Shan, R. Miura, T. Kagawa,[7] presented a research paper on unmanned aerial vehicle (UAV), also called a drone, is getting more and more important in many emerging technology areas. For communication area, the drone also takes an important role in lots of significant topics like emergency communications, device-to-device (D2D) communications, and the Internet of Things (IoT). One of the important drone applications is to collect and share data among drones and other aircraft, which is useful for drone control so that dangerous conditions can be avoided. In particular, the drone control and safety guarantees are difficult to attain, especially, when drones fly beyond the line of sight (BLOS). For this reason, we develop a drone location information sharing system using the 920-MHz band. We use this system to do a long distance propagation field experiment for model establishment. Unfortunately, the current data collection for model establishment work needs a great effort and time to do experiments to collect a huge number of data for data analysis so that a suitable model can be established. Therefore, in this paper, we propose a novel method, which is based on machine learning approach, to data analysis and model establishment for drone communications, so that the effort and cost for establishing model can be reduced and a model, which captures more details about the drone communications, can be obtained. The results of this paper validate that the proposed method can indeed establish a more complicated model with less effort. Specifically, from the distribution of the training error, it can be known that there are over 80% training errors with intensity less than 5, which ensures the error performance of the proposed method.

1.3 Advanced implementation in drone for Smart City Developments using ML Algorithm

Sampath Boopathi [8] presented a chapter explores the integration of drones, machine learning, and artificial intelligence (AI) in smart city development. Drones can revolutionize urban planning, energy efficiency, noise reduction, environmental monitoring, traffic management, infrastructure inspection, public safety, data security, and privacy protection. AI-driven solutions enable data-driven decision-making for resource allocation, sustainability, and predictive modeling. AI optimizes flight paths for energy efficiency, noise reduction strategies enhance drone social acceptance, and autonomous drone navigation is crucial for safe urban deployment. Drones optimize traffic flow, reduce congestion, and enhance safety. AI provides predictive insights into traffic patterns, and drones aid in law enforcement, emergency response, and first responder support. Data security and privacy protection measures are essential for maintaining public trust.

1.4 Sound-Based Amateur Drone Detection for Public Safety Applications

Muhammad Zohaib Anwar,[9] presented that In recent years, popularity of unmanned air vehicles enormously increased due to their autonomous moving capability and applications in various domains. This also results in some serious security threats, that needs proper investigation and timely detection of the amateur drones (ADr) to protect the security sensitive institutions. In this paper, we propose the novel machine learning (ML) framework for detection and classification of ADr sounds out of the various sounds like bird, airplanes, and thunderstorm in the noisy environment. To extract the necessary features from ADr sound, Mel frequency cepstral coefficients (MFCC), and linear predictive cepstral coefficients (LPCC) feature extraction techniques are implemented. After feature extraction, support vector machines (SVM) with various kernels are adopted to accurately classify these sounds. The experimental results verify that SVM cubic kernel with MFCC outperform LPCC method by achieving around 96.7% accuracy for ADr detection. Moreover, the results verified that the proposed ML scheme has more than 17% detection accuracy, compared with correlation-based drone sound detection scheme that ignores ML prediction.

1.5: Reinforcement learning framework for UAV SAR For optimization of path

Ramezani and Amiri Atashgah [10] displayed an energy-aware hierarchical reinforcement learning (HRL) framework for unmanned air vehicle (UAV) search and rescue missions that integrate LSTM-based battery consumption prediction for optimizing path planning. Their framework involves a high-level controller (HLC) that sets feasible goals based on predicted battery requirements and a low-level controller (LLC) that executes navigation commands. A novel innovation is using a bidirectional LSTM model to accurately predict battery consumption for different targets, helping the HLC make energy-efficient decisions. The framework includes hindsight experience replay (HER) to improve learning efficiency and adaptability. The system was validated through extensive simulations in MATLAB using four test environments with varying obstacle densities and target counts. Results demonstrated superior performance compared to baseline methods, achieving 90-95% success rates across environments while maintaining low collision rates of 3-8%. The authors' hierarchical approach with LSTM battery prediction enabled more efficient path planning and energy utilization compared to traditional reinforcement learning methods. Key advantages included setting achievable goals based on energy constraints and

adapting to dynamic environments while optimizing battery usage through predictive insights. Test results verified the framework's effectiveness for autonomous UAV navigation in complex search and rescue scenarios.

1.6: Deep learning approach in a multi-obstacle environment.

In this study, Zhang et al. [11] displayed a deep reinforcement learning (DRL) method for autonomous UAV navigation in dynamic multi-obstacle environments. The key methodological contributions include 1) the Development of a two-stream Actor-Critic network structure based on Twin Delayed Deep Deterministic Policy Gradients (TD3) that considers both current observations and changes in environment observations to predict obstacle movements, 2) the Implementation of a Unity3D simulation platform to reduce the reality gap, and 3) Design of a comprehensive reward function incorporating distance, step penalty, angle penalty and obstacle avoidance components. The method was evaluated through extensive experiments in both static and dynamic environments with varying obstacle configurations. Results demonstrated superior performance compared to baseline methods:

- In static environments with random obstacles, their method achieved up to 95.25% success rate compared to 92.10% for DDPG and 95.15% for conventional TD3
- In dynamic environments with moving obstacles, their approach maintained a 77.65% success rate even at high obstacle speeds (20 m/s), while DDPG dropped to 59.80%
- The method showed better generalization ability and faster convergence during training compared to baselines.
- The two-stream network structure enabled effective prediction of obstacle movements, allowing proactive collision avoidance.
- Ablation studies on reward function design validated their formulation's ability to balance safety and efficiency

1.7: Air-to-Air Simulated Drone Dataset for AI-powered problems

The Air-to-Air Simulated Drone Dataset (A2A-SDD) [7] significantly contributes to counter-UAV research through its comprehensive simulation-based approach to data generation [7]. Using the AirSim platform, this dataset provides diverse drone interaction scenarios, featuring up to three monitoring drones pursuing one or two target drones in various environmental conditions [7]. The

dataset's architecture incorporates five different drone types, including several DJI models (Phantom, Mavic, FPV, and Inspire) alongside a generic quadrotor model, offering a broad representation of common UAV platforms [7].

A key innovation of A2A-SDD lies in its multi-view data collection methodology [7]. The dataset provides richer spatial information by capturing scenarios from multiple camera perspectives than traditional single-view approaches. This multi-perspective design is complemented by comprehensive annotations, including precise XYZ coordinates for all drones, detailed bounding box information, and payload status indicators [7]. The dataset also incorporates various environmental conditions, simulating sunny, rainy, and snowy weather across different terrains, from natural landscapes to urban environments [7].

The practical utility of A2A-SDD has been demonstrated through experimental validation using deep learning models [7]. The researchers achieved notable results in two critical areas: distance estimation and payload detection. The distance estimation models achieved an RMSE of approximately 8.4 meters when tested on real-world data, while payload detection systems demonstrated accuracies between 87-92% in practical applications [7]. These results validate the dataset's effectiveness in bridging the gap between simulated training data and real-world applications.

However, A2A-SDD does face certain limitations [7]. The reliance on simulation, while enabling comprehensive data collection and annotation, raises questions about fully representing real-world complexities. The current focus on visual data also allows expansion into other sensing modalities [7]. Despite these limitations, the dataset is valuable for developing and testing counter-UAV measures, particularly in scenarios where real-world data collection would be impractical or cost-prohibitive.

The development of A2A-SDD marks a significant step forward in counter-UAV research, providing researchers with a robust platform for developing and testing detection, tracking, and classification algorithms [7]. The dataset's comprehensive nature, combined with its validated effectiveness in real-world applications, establishes it as a valuable resource for advancing the field of drone detection and counter-measure development.

1.8: A mixed deep learning approach for planning UAV routes in changing conditions

Zhang [13] presented a paper with the use of Unmanned Aerial Vehicles (UAVs) in military operations has grown in importance. UAVs are typically operated in uncertain environments during actual missions, necessitating careful planning to adjust in real time. For UAVs to navigate through challenging terrain, an efficient path-planning algorithm is also necessary. But for UAV path planning, the widely utilized Rapidly-exploring Random Tree (RRT) method frequently produces less-than-ideal routes that need a lot of post-processing to make them better. Its variations have higher memory and compute costs, but they can converge to the best answer. However, purely learning-based approaches could lead to poor planning or overly lengthy courses created in order to avoid obstacles. In this study, we provide a neural network architecture including CNN, LSTM, and an attention mechanism for UAV path planning that combines a variety of sensor inputs. The neural module and informed sampling techniques are then combined by the deep learning-enhanced RRT algorithm to provide adaptive and effective pathfinding in dynamic contexts. Our method outperforms the state-of-the-art Layered RQN, Informed RRT*, and Batch Informed Trees (BIT*) algorithms in terms of overall performance after carefully weighing the trade-offs between path optimality, computing efficiency, and fewer turning angles.

1.9 Using multi-agent reinforcement learning for drone mapping

Riccardo Zanoli [14] presented a paper explaining drone mapping and surveying for objects of interest, such forest fires, landslides, or wild animals, has become more popular in recent years across a variety of academic communities. However, the usefulness of the drones is significantly limited by the requirement for a human pilot or a pre-planned flight path, particularly when an entire swarm is deployed. In order to demonstrate that the performance loss resulting from the absence of explicit optimization and pre-programmed knowledge of the system statistics is minimal in the swarm scenario, we present a model of the drone survey problem and apply three well-known reinforcement learning techniques.

Section 2: A-Star Algorithm for drone system

2.1: UAV trajectory optimization using the A-star algorithm and clustering-based strategic 3D deployment for improved disaster response.

Humairah Hamid [13] presented a broad spectrum of communication and information technologies is currently being investigated for their potential applications in disaster management. A high level of situational awareness, combined with a prompt and accurate response, is essential for the preservation of life during catastrophe scenarios. This study presents a novel communication strategy employing Unmanned Aerial Vehicles (UAVs) as aerial base stations for providing connectivity to the affected area. The system takes advantage of the flexibility and quick deployment characteristics of UAVs. The main focus is to determine the optimal UAV deployment along with trajectory planning to ensure connectivity in areas where conventional base stations are inaccessible. The proposed system employs two types of UAVs: cluster UAVs which act as stationary base stations and relay UAVs acting as mobile base stations. A three-step strategy is proposed to find the suitable location of cluster UAVs, optimize their height and power, and find the optimal trajectory of the relay UAVs to maximize the percentage of users served. Gaussian Mixture Model (GMM) clustering is employed to determine the optimal horizontal location of cluster UAVs. An optimization problem is framed for finding out the optimal height and power for cluster UAVs. Heuristic-based A-star algorithm is used to find out the trajectory of the relay UAVs which can efficiently minimize the overall path length while avoiding obstacles. The simulation results confirm the effectiveness of the proposed approach and demonstrate performance enhancement by comparing it with the benchmark schemes.

2.2 Improved method using A-Star algorithm for model R5DOS

Based on the R5DOS model,[14] this paper suggests a three-dimensional space UAV path planning model to address the issues of the A-star algorithm's lengthy computation times and massive computation volumes in three-dimensional space. Together, the enhanced A-star algorithm and the improved R5DOS intersection model are used. They build a local search process together, and by lowering the number of search nodes, the R5DOS path planning model is established. By simulating the path planning model with MATLAB software, the model can significantly lower the computational complexity and number of nodes of the A-star algorithm in three-dimensional spaces, as well as the UAV's calculation time. Lastly, we contrast the

refinement of the A-star algorithm with both the original and geometric versions. It is demonstrated by the final fitting result that the enhanced A-star algorithm requires fewer node visits and has a shorter computation time. In general, the simulation results validate the efficacy of the enhanced A-star algorithm and can serve as a guide for further path planning algorithm research.

2.3 Optimal route planning for stealth unmanned aerial vehicles in a 3D network radar environment using a modified version of the A-Star algorithm.

Zhe Zhang [15], presenter a paper on Path security and search efficiency of penetration paths are the two most crucial aspects of mission execution for stealth unmanned aerial vehicles (UAVs). The optimal penetration path planning technique that takes into account the network radar system, the dynamic radar cross-section of stealth UAVs, and kinematics principles is examined in this article. The traditional A-Star algorithm was modified to include a 3D bidirectional sector multilayer variable step search strategy and the radar threat estimation function. This modified A-Star algorithm seeks to meet both the algorithm searching efficiency and waypoint accuracy requirements. Attitude angle constraints and the radar system's rank-K fusion criterion were then met by the simultaneous selection of new waypoints using the suggested penetration path planning method. The conventional A-Star algorithm, bidirectional multilayer A-Star algorithm, and modified A-Star algorithm were also used for comparative analysis of various algorithms in order to resolve the penetration path issue that UAVs encounter under different threat scenarios. In a 3D complex network radar environment, the simulation results demonstrate the efficacy of the suggested path planning scheme by showing that the paths produced by using the modified algorithm have optimal path costs and higher safety.

Section 3: Search and Detection Operation in drone system

3.1: Path planning algorithm based on Bayesian Inference

M. Khosravi et al. [16] presented a comprehensive autonomous drone system for search and detection operations, specifically covering search and rescue (SAR) and wildfire detection scenarios. Their approach integrates two key components: a path-planning algorithm based on Bayesian inference and a target detection system using a residual neural network (ResNet). The path planning algorithm optimizes the drone's route by utilizing probabilistic knowledge of target locations, while the ResNet-based detection system achieves more than 91% accuracy in

identifying targets from aerial imagery. The system employs a sliding window technique to manage computational complexity and incorporates probability updating through Bayesian inference for improved search efficiency. Their system demonstrated superior performance through simulation and real-world experiments compared to benchmark algorithms, achieving significantly reduced search times and energy consumption. Their approach proved five times faster than traditional zigzag search patterns for a Gaussian mixture distribution scenario. The authors evaluated their system by using a DJI Mavic 2 Zoom drone equipped with a 4K camera, testing it in various conditions involving fire detection and person detection in snowy environments. While proving effective in these different scenarios, the authors note potential improvements through ensemble ML models and extension to multi-UAV operations.

3.2 SAR Operations in disaster environment with bio radars.

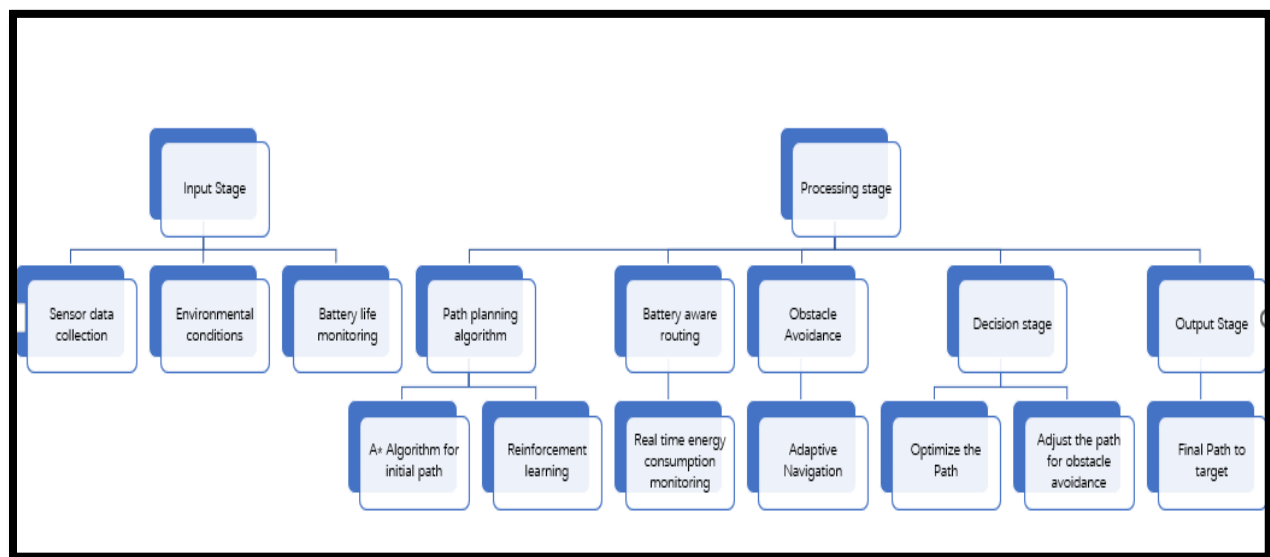
Mittal et al. [17] presented a vision-based autonomous UAV navigation and landing system designed explicitly for Urban Search and Rescue (USAR) operations in post-disaster environments. The system enables UAVs with bio radars to autonomously navigate through and land on collapsed buildings to locate survivors. Their key contributions include (1) A novel landing site detection algorithm that evaluates multiple hazard factors, including terrain flatness, steepness, depth accuracy, and energy consumption through weighted cost maps; (2) An efficient exploration strategy using a next-best-view planner combined with TSDF-based 3D reconstruction for mapping disaster environments; and (3) A minimum-jerk trajectory generator for safe landing maneuvers. The system ran in real time on an embedded computer. It was extensively validated both in simulation using their introduced synthetic dataset of 1.2 million images and in real-world experiments at a rescue training facility. Results demonstrated reliable detection of safe landing sites on rubble piles and debris, outperforming existing approaches that tend to favor flat ground surfaces. The authors also released the first large-scale dataset of collapsed buildings with ground truth depth, surface normal, and semantic labels to facilitate further research.

3.3 Autonomous Motion Planning System for Outdoor Environments with Uncertainty in Object Detection Using Drones.

Sandino et al. [18] presented a modular system for autonomous motion planning and object finding using small UAVs, focusing on search and rescue (SAR) missions. The system employs a Partially Observable Markov Decision Process (POMDP) to enhance UAV autonomy for obstacle

avoidance, environment exploration, and object inspection under uncertainty conditions. The architecture integrates Software-in-the-Loop (SIL) for PX4 autopilot emulation and Hardware-in-the-Loop (HIL) for onboard inference using a MobileNet SSD detector and motion policy computation via TAPIR and ABT solver. The system was evaluated in simulation and preliminary real flights using three flight modes: mission (baseline), offboard (POMDP-based), and hybrid (combined approach). Testing involved locating a victim in two scenarios with varying occlusion levels. Results showed offboard and hybrid modes achieved superior detection accuracy (100% and 92.5% respectively) compared to the baseline, with minimal collision rates (1.25% overall) during navigation. The research notably demonstrated improved robustness against false positives and the ability to handle object detection uncertainty in outdoor environments using resource-constrained hardware, though flight times increased due to the comprehensive inspection approach. The study recommends future work including detailed real-flight analysis, evaluation with additional sensors like thermal cameras, and investigation of more complex scenarios with multiple or dynamic targets.

3. Research Program



3.1 Research Objective and content

This research aims to develop and implement an advanced path planning and optimization system that combines A* path finding algorithm with machine learning reinforcement learning, which is a method that would enhance efficiency, adaptability and the capabilities of the drone. The following objectives and contents outline the key areas of investigation:

1. Path planning algorithm based on A* and reinforcement learning

- Design and implement a path planning approach combining the algorithm A* with AI module.
- Create an environment for testing and validation.
- Integrating the necessary inputs for the environmental perception.
- Enable dynamic path recalculations when new obstacles appear in the route.
- Implementing battery consumption prediction and monitoring throughout the route for battery-efficient coverage routes.

2. Optimization through reinforcement learning approach

- Enhance path planning efficiency using machine learning algorithms.
- Optimize drone battery efficiency and mission completion time.
- Develop an obstacle avoidance approach for dynamic environments.

3. adaptive navigation environment modeling

- the target position.
- complex obstacles.
- dynamic environment changes.

4. Performance analysis and validation

- Evaluate the effectiveness of the ML approach.
- Analyze system performance in various environmental conditions.
- Real-time computational performance.

3.2 Key Research Problems

1. Real-time Path planning challenges

- Dynamic obstacle avoidance
- Computational efficiency in complex environments
- Implementation of the ML approach

2. Optimization Issues

- Battery efficiency optimization
- Mission time reduction
- Path selection optimization under different conditions

3.3 Methods

1. Input Stage

The input stage involves collecting real-time data from various sources that influence the drone's navigation and mission execution. This data forms the foundation for intelligent path planning and optimization. Key inputs include:

- Simulated Sensor Data Collection

Simulated Sensors on the drone collect real-time data about the environment, including terrain structure, weather conditions, and potential obstacles.

Simulated Sensors include LiDAR, ultrasonic sensors, GPS, and inertial measurement units (IMUs) to provide accurate spatial positioning and obstacle detection.

- Environmental Conditions

Static obstacles: Buildings, trees, and fixed structures.

Dynamic obstacles: Moving vehicles, people, and other drones.

Weather factors: Wind speed, rainfall, and temperature variations can affect drone stability and flight path.

- **Battery Level Monitoring**

Monitoring real-time battery consumption helps ensure the drone does not run out of power mid-mission.

Battery-aware routing ensures that the drone can return to the base or complete its mission before depleting the battery.

2. Processing Stage

The processing stage is where the drone processes the collected data to compute an optimized path to the target location. It includes the following key components:

Path Planning Algorithm

1. **A* algorithm**

Used for initial path calculation.

Finds the shortest and most efficient path considering static obstacles.

2. **Reinforcement Learning (RL):**

RL enables the drone to adapt to dynamic changes in the environment.

The algorithm improves over time by learning from previous navigation data and adjusting to new obstacles.

Battery-Aware Routing

The drone constantly evaluates its battery consumption and adjusts the route if the remaining battery is insufficient to complete the mission.

The algorithm balances between finding the shortest path and conserving battery life.

Obstacle Avoidance

The drone uses real-time sensor data to detect obstacles.

If an obstacle appears on the planned path, the RL-based system recalculates the route to avoid collisions while maintaining mission efficiency.

Reinforcement Learning Technique in the method

- **Intelligent Path Planning:** Use RL agents trained to learn optimal navigation strategies in an environment modeled with various obstacles, terrain types, and drone constraints (like battery).
- **Dynamic Obstacle Avoidance:** Implement Deep Q-Learning (DQN) or Proximal Policy Optimization (PPO) for training the drone to avoid obstacles that appear dynamically.
- **Battery-Efficient Routing:** Incorporate battery level as a state feature in your RL environment. Design a reward function that penalizes high energy consumption or risks of not returning to base.

Path Optimization via RL

Model-Based RL

Model-Based RL is a type of reinforcement learning where the agent learns a model of the environment's dynamics—that is, it tries to predict what will happen next (next state and reward) given the current state and action. Once the model is learned, the agent uses it to plan ahead and make better decisions.

Usage of Model-Based RL for Drone Path Planning

Drones operate in complex, dynamic, real-time environments, and Model-Based RL offers several advantages for your research:

1. Sample Efficiency

- Fewer real-world/simulation interactions are needed to learn good policies.
- Critical when simulations are costly or time-consuming.

2. Faster Adaptation

- The learned model can simulate future trajectories and outcomes quickly.
- Help in anticipating dynamic obstacles or planning battery-constrained routes.

3. Better Planning Capabilities

- The drone can "imagine" several future possibilities and choose the best one.
- Ideal for urban search and rescue, where the environment is unpredictable.

Technique of execution:

- Create an internal model of the environment and simulate future outcomes.
- Useful when simulating drone physics, energy use, or environmental uncertainty.

3. Decision Stage

The decision involves dynamically adjusting the path based on real-time environmental changes and resource constraints:

- Optimize Path Based on Dynamic Conditions

The algorithm continuously evaluates the environment and updates the path to avoid unexpected obstacles.

- Battery levels, weather conditions, and external disturbances are factored into real-time decision-making.

Adjust Path for Obstacle Avoidance and Battery Constraints

If the calculated path becomes inefficient due to a new obstacle or low battery, the system recalculates the path dynamically.

The goal is to reach the target location efficiently without compromising the drone's safety and mission success.

4. Output Stage

The output stage presents the final path and performance metrics, which provide insights into the drone's mission execution:

- Final Path to Target

The final path is optimized for the shortest distance, minimal energy consumption, and minimal risk of collision.

The drone can adapt to real-time environmental changes while maintaining accuracy and efficiency.

- Performance Metrics

Path Efficiency – Measures how close the calculated path is to the shortest possible route.

Collision Rate – Tracks the drone's ability to avoid obstacles in real time.

Battery Usage – Measures the efficiency of energy consumption relative to the distance traveled and environmental complexity.

3.4 Novelty

This research aims to create a novel hybrid architecture that integrates the A* algorithm with the reinforcement learning to create an adaptive and efficient path planning and optimization system for drone navigation. The innovation lies in combining the A* efficiency with the optimization of the machine learning, in addition to incorporating intelligent battery-efficient methods through predictive consumption modeling and dynamic route optimization. The system provide real-time environmental adaptation capabilities, including dynamic obstacle avoidance and path reconfiguration, all within a scalable framework. The proposed architecture implements a novel reward function design that balances multiple objectives while maintaining computational efficiency for real-time operation. This approach is different than the current approaches by providing integrated system that addresses the effectiveness and efficiency of both the path finding and resource optimization in the same framework, making this approach suitable for complex urban search and rescue missions.

4. Expected Objectives and Achievements

1. Expected research objectives:

1. Intelligent path planning and optimization algorithm

- Develop a novel drone pathfinding and optimization system that integrates the machine learning technique with the classical path-finding A* algorithm.

2. Dynamic obstacle Avoidance

- Design an algorithm that can minimize collision risks while maintaining mission efficiency.

3. Battery-Efficient Routing

- Implementing optimization strategies that reduce unnecessary drone movements and calculate the shortest routes to the target.

Selection of environment	•The suitability of interphase with creative ideas on the display
Creation of dataset	•Real time data gathering with onboard sensors like Lidar, GPU, Ultrasonic to detect obstacles and environmental changes.
Data filtration	•Removal of null values ore wrong values.
Implementation of Algorithm	•Development of supervised technique for initial path planning and then reinforcement learning for adaptive learning.
Accuracy and precision	•Performance evaluation based on factors like path accuracy , collision rate and battery performance.
Comparison with previous techniques	•Comparative study on the previous similar work.

Project Flow Chart

The image outlines a structured framework for intelligent path planning using AI and reinforcement learning, likely for an Urban Search and Rescue (SAR) mission with a single drone. detailed explanation of each stage:

1. Selection of Environment

- **Objective:** Identify and define the environment where the path planning will be executed.
- **Key Considerations:**
 - **Suitability:** The environment should support the interfacing and visualization of the path planning algorithm.

- **Creative Display:** The interface should present data and results in an intuitive and creative way, enhancing the operator's situational awareness.
- **Challenges:** Urban environments may have dynamic obstacles, narrow paths, and unpredictable changes in terrain or structure.

2. Creation of Dataset

- **Objective:** Collect real-time data from sensors mounted on the drone.
- **Data Sources:**
 - **Lidar:** Provides high-resolution mapping of the surroundings, detecting obstacles and terrain variations.
 - **GPU:** Processes complex data quickly for real-time decision-making.
 - **Ultrasonic Sensors:** Detect nearby objects and measure distances with high accuracy.
- **Outcome:** A comprehensive dataset that reflects the real-time status of the environment and potential obstacles.

3. Data Filtration

- **Objective:** Clean and preprocess the collected data.
- **Process:**
 - **Remove Null Values:** Missing data points are removed to prevent processing errors.
 - **Correct Errors:** Wrong or inconsistent data values are filtered out.
 - **Noise Reduction:** Data from sensors can have noise due to environmental interference, which needs to be corrected.
- **Outcome:** A clean, structured dataset ready for algorithm training and execution.

4. Implementation of Algorithm

- Objective: Develop and apply an AI-based path planning algorithm.
- Methodology:
 - Reinforcement Learning: Applied for adaptive learning where the algorithm continuously improves based on real-time feedback.
 - Dynamic Path Adjustment: The algorithm recalculates the path dynamically in response to environmental changes.
- Outcome: A robust and adaptive path planning solution capable of handling real-time changes.

5. Accuracy and Precision

- Objective: Evaluate the performance of the path planning algorithm.
- Key Metrics:
 - Path Accuracy: How close the planned path is to the optimal path.
 - Collision Rate: Frequency of the drone hitting obstacles.
 - Battery Performance: The efficiency of path planning in terms of energy consumption.
- Outcome: Quantitative insights into the algorithm's effectiveness and areas for improvement.

6. Comparison with Previous Techniques

- Objective: Benchmark the developed algorithm against existing methods.
- Key Factors:
 - Performance: How well the new algorithm outperforms previous solutions in terms of accuracy, speed, and energy consumption.
 - Adaptability: How effectively the algorithm handles dynamic changes compared to traditional methods.

- Efficiency: Reduction in computational complexity and resource consumption.
- Outcome: Validation of algorithm improvements and unique advantages over previous techniques.

2. Expected Achievements:

1. Technical innovation

- A hybrid path planning system that successfully combines A* algorithm with machine learning techniques

2. Performance metrics

- Demonstrate significant improvements in path efficiency, collision avoidance, and adaptability.

3. adaptability

- Prove that the system is adaptable through different urban SAR scenario types.

5. Research plan

MONTH	ACTIVITIES	DELIVERABLES
October 2024	Literature review- Research methodology planning – Python and ML review	Documents
November 2024	Thesis proposal - writing -initial tools and libraries setup	The updated versions
December 2024	Thesis proposal writing and submitting	The updated versions
January 2025	Test Environment setup	The updated versions
February 2025	Review of implementation against requirement- technical documentation	The updated versions
March 2025	Path planning and optimization algorithm implementation- mid-term assessment preparation- code review and documentation	The updated versions
April 2025	Algorithm completion- comprehensive testing documentation- results analysis and optimization	The updated versions
May 2025- July 2025	ML implementation and analysis – dataset documentation	The updated versions
August 2025	Performance review- system optimization – performance documentation	The updated versions

September 2025	Integration testing – documentation – result compilation	The updated versions
October 2025	Final testing	The updated versions
November 2025	Thesis finalizing- defense preparation	The updated versions
December 2025	Final review and updates – defense presentation preparation	The updated versions

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